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Determinants of Foreign exchange exposure during the COVID-19 crisis: A Multi-country evidence

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ABSTRACT

The research question of which firm-level factors make firms more vulnerable to exchange rate fluctuations during periods of crisis has rarely been explored by prior literature. Using a large sample of 1577 firms from 9 developed and 11 emerging countries, this study presents a comprehensive analysis of how firm-level factors affect firms' foreign exchange exposure before and during the COVID-19 crisis. The results provide evidence of a substantial increase in firms' linear exposure during the COVID-19 period. The cross-sectional analysis reveals that the effects of firm-level variables on exposure are more pronounced during crisis periods and are different from non-crisis periods. Firms that have effective asset utilization or large operating profit margins remain less exposed during times of stress. Contrary to hedging theory, firms that have high incentives to hedge such as firms with high financial leverage become highly exposed to currency fluctuations during crisis periods. The interaction analysis provides further evidence that firms with high leverage can limit their foreign exchange exposure during periods of crisis if they have high asset turnover or high operating profits. The results offer important practical implications to firms for risk management during periods of crisis.

KEYWORDS

Foreign exchange exposure; COVID-19; firm-level; forex risk; crisis

I. Introduction

Foreign exchange (forex) exposure of a firm is the sensitivity of its cash flows to the unanticipated changes in exchange rates (Adler and Dumas 1984). Although there is ample evidence that a firm's forex exposure is determined by its international activities (Jorion 1990; Choi and Prasad 1995), size (Dominguez and Tesar 2006), hedging incentives (He and Ng 1998; Allayannis and Ofek 2001), and competition (Williamson 2001), little is known about the role of firm-level fundamental factors in explaining its forex exposure, particularly during periods of financial crisis. It is imperative for firms to know what specific factors make them more vulnerable to exchange rate fluctuations during periods of crisis. Prior studies that analyse the firms' forex exposure during crisis periods are limited (Aquino 2005; Muller and Verschoor 2007) and have not explored the relationship between firm-level factors and forex exposure during times of stress.

This study addresses this gap by exploring the research question: How does the within-firm relationship between forex exposure and its firm-level determinants change during periods of financial crisis. Specifically, this study conducts a comprehensive investigation of how firm-level factors affect forex exposure before and during the COVID-19 crisis.

An in-depth analysis of firms' forex exposure in the context of COVID-19 could provide useful and interesting insights to the firms for several reasons. Firstly, COVID-19 is declared unique by economists and radically different from the previous financial crises due to its nature of being truly exogenous, uncertain, and global (Borio 2020). Thus, the implications of prior studies on risk management that focus on various global crises (Aquino 2005; Verschoor and Muller 2007) are hardly applicable in the current context. Secondly, a significant rise in the exchange rate volatility was observed in world currency markets during the COVID-19 pandemic (Table 1) which indeed raises concerns about firms' high exposure to foreign exchange rate risk. Lastly, with regard to the current literature on COVID-19, many studies have explored the effect of COVID-19 on stock market risk (Baek and Lee 2021), systemic risk (Rizwan, Ahmad, and Ashraf 2020), and oil price risk exposure (Akhtaruzzaman et al. 2020),

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Table 1. Exchange rate volatility across countries.

Country	Annualized volatility of the exchange rate index (Pre-COVID -19 period)	Annualized volatility of the exchange rate index (COVID-19 period)	Increase (+)/decrease(-) in the volatility during the COVID-19 period
Developed			
Canada	1.29%	1.46%	0.17%
France	0.39%	0.55%	0.16%
Germany	0.50%	0.73%	0.24%
Italy	0.44%	0.64%	0.20%
Japan	1.42%	2.14%	0.72%
Spain	0.37%	0.54%	0.17%
Sweden	1.19%	1.29%	0.10%
UK	1.44%	1.83%	0.39%
US	0.95%	1.32%	0.36%
Average	0.89%	1.17%	0.28%
Developing			
Brazil	2.55%	4.24%	1.69%
Chile	1.98%	3.01%	1.04%
China	0.78%	0.93%	0.15%
Colombia	2.09%	3.70%	1.61%
India	1.08%	1.15%	0.07%
Mexico	2.25%	4.33%	2.09%
Peru	0.85%	1.38%	0.53%
Philippines	0.94%	1.04%	0.10%
Poland	0.89%	1.23%	0.34%
South Africa	2.85%	3.27%	0.42%
Turkey	1.81%	3.88%	2.08%
Average	1.83%	2.37%	0.54%

This table compares the annualized volatility of the nominal exchange rate indices across countries during the pre-COVID-19 and COVID-19 periods. The pre-COVID-19 period is from Jan 2017 till the date of the first reported case for each country. The COVID-19 period ends on the day when the vaccination program begins in the country.

however, to the best of our knowledge, no study has analysed the determinants of forex exposure in a comprehensive manner during the COVID-19 pandemic.

Following prior literature,[1] this study employs a two-stage methodology in which, in the first stage, the firm-specific forex exposure is estimated using an augmented market model[2] (Parsley and Popper 2006), while in the second stage, the relationship between exposure and firm-level factors such as size, asset turnover, profitability and proxies for hedging incentives are explored using the cross-sectional regression. Using a sample of 1577 firms from 20 countries (9 developed and 11 developing countries), the analysis is performed for two time periods- the pre-COVID-19 period and the COVID-19 period.

The study finds that the COVID-19 period witnessed significantly higher forex exposure of sample firms across all the industries in terms of prevalence and magnitude. The findings support a stronger presence of linear exposure than asymmetric or non-linear exposure.

During the COVID-19 period, the results show that firms with effective asset utilization or high profitability are less exposed to exchange rate fluctuations. Similarly, consistent with hedging theories (Smith and Stulz 1985), firms that have high incentives to hedge such as firms with high leverage or low liquidity are less exposed. However, despite high incentives to hedge, high-growth firms may prioritize market share objectives over forex risk management (Krapl 2017) and thereby are exposed more. The findings are mostly similar for the two sub-samples of firms from developed and developing countries except for firm size. Firm size has a negative effect on exposure for firms from developed countries indicating that larger firms are more motivated to hedge due to the advantage of economies of scale in their hedging activities (Nance et al. 1993). On the contrary, size is positively related with exposure for firms in developing countries indicating that smaller firms in these countries hedge more to reduce their high financial distress costs and thus are exposed less.

Further analysis reveals that the effects of firm-level variables on forex exposure are more pronounced during the COVID-19 period and are different, specifically for firms with high leverage. Firms with high financial leverage become significantly highly exposed to forex risk during the period of the COVID-19 crisis as they might be less able to hedge due to high hedging costs (Wei and Starks 2013). The interaction analysis provides evidence that firms with high leverage can limit their forex exposure during the period of the COVID-19 if they have high asset turnover or high operating profits. These results are robust to the dynamic panel GMM estimation.

The study makes the following contributions to the body of literature. First, it improves our understanding of the firm-level determinants of forex exposure during periods of crisis which may have important implications for firms in limiting their forex risk. To the best of our knowledge, it is the first of its kind that

[1]Aggarwal and Harper (2010); Choi and Prasad (1995); Chue and Cook (2008); Dominguez and Tesar (2006); Francis et al. (2017); He, Liu, and Zhang (2021); He and Ng (1998); Hutson and Laing (2014); Hutson and Stevenson (2010); Loriot, Hutson, and Au Yong (2020) and many others.
[2]An extension of the standard Jorion's (1990) model.

presents a comprehensive analysis of firms' forex exposure in the dynamic context of COVID-19. Second, the findings provide novel evidence that the relationship between firm-level variables and exposure became stronger or changed during periods of crisis which offers useful insights for firms for effective risk management. Third, it identifies how firms with high leverage can limit their exposure to currency fluctuations during times of crisis. Finally, using a larger dataset than those of prior studies – covering 1,577 firms from 20 countries, it provides a comparison between the firms from developed countries and developing countries. The findings offer important practical implications to firms for risk management during crisis periods.

The rest of the paper is summarized as follows. Section II presents a brief review of the existing literature. Section III explains the data, sample selection, and methodology followed by the empirical results in section IV. Finally, section V provides the implications and concluding remarks.

II. Literature review

This section presents a summary of the existing literature that analyzes forex exposure and its determinants.

Determinants of forex exposure

Firm-level factors identified by prior literature that explain the cross-sectional variations in forex exposure are firms' international activities, size, hedging incentives, profitability, industry structure, and competition (Jorion 1990; He and Ng 1998; Williamson 2001; Dominguez and Tesar 2006). Table 2 provides a summary of important firm-level determinants, their theoretical relationship with exposure, and empirical support.

Size

Theories that explain the relationship between firm size and exposure provide contrasting views. As per optimal hedging theories which state that hedging reduces the costs of financial distress (Smith and Stulz 1985), smaller firms have higher incentives to hedge due to their higher financial distress costs and thus, are less exposed to forex risk. On the other hand, larger firms are more motivated to hedge due to the advantage of economies of scale in their hedging activities (Nance et al. 1993), and therefore, should have lower exposure. Prior empirical studies have found a mixed relationship between firm size and forex exposure (Bodnar and Wong 2003; Dominguez and Tesar 2006; Loriot, Hutson, and Au Yong 2020).

Table 2. Firm-Level determinants of forex exposure.

Variable	Variable's predicted relationship with exposure	Supported literature	Theoretical Motivation
Firm Size	±	Chow et al. (1998); Bodnar and Wong (2003); Dominguez and Tesar (2006); Chang et al. (2013), Chue and Cook (2008); Loriot, Hutson, and Au Yong (2020)	Larger firms are more likely to hedge due to the advantage of economies of scale in their hedging activities and should have lower exposure than smaller firms. On the contrary, smaller firms have a higher incentive to hedge due to their higher financial distress costs and should be less exposed to forex risk
Growth opportunities	-	Froot, Scharfstein, and Stein (1993); He and Ng (1998); Nance et al. (1993)	Firms with high growth opportunities have more incentives to hedge to reduce the underinvestment problem and thus have lower exposure
Leverage	±	Smith and Stulz (1985); He and Ng (1998); Nance et al. (1993)	Firms with high costs of financial distress hedge more and have less forex exposure
Liquidity	+	Smith and Stulz (1985); He and Ng (1998); Nance et al. (1993)	Firms with higher liquidity hedge less due to less financial distress costs and have higher exposure
Efficiency	-	Aggarwal and Harper (2010)	Firms with higher asset turnover are able to adjust to the dynamic competitive environment and are less exposed
Profitability	-	Carlos, Sophie, and Tang (2018); Muller & Verschoor, 2006	Firms with high profitability have high cushions to bear hedging costs and are less exposed
Openness	+	Hutson and Stevenson (2010); Chue and Cook (2008)	Higher involvement of the country in international activities leads to higher exposure
Industry	±	Allayannis and Ihrig (2001); Williamson (2001)	Industries differ in terms of pass-through and mark-ups, competitive structure, or industry concentration and hence may face different levels of exposure

This table reports the industry classification of sample firms for each country. Tel. is Telecommunications, Tech. is Technology, CS is Consumer services and CG is Consumer Goods.

Variables as proxies for the hedging incentives

The most important determinant of forex exposure is hedging. The negative relationship between hedging and exposure is widely accepted by theoretical and empirical literature (Allayannis and Ofek 2001). Due to the non-availability and incompleteness of data on firm-level hedging activities, consistent with prior literature (Chow and Chen 1998; He and Ng 1998; Muller and Verschoor 2007; Hutson and Stevenson 2010), variables that proxy for hedging incentives are used to determine the relationship between hedging and exposure. This literature assumes that firms are motivated to hedge if hedging benefits are greater than costs and the incentives to hedge are positively related to the likelihood or effectiveness of hedging (Nance et al. 1993). Therefore, firms with higher hedging incentives would hedge more and consequently, would have lower forex exposure. Three variables are used in this study as proxies for the hedging incentives- financial leverage, growth opportunities, and liquidity.

Since optimal hedging theories postulate that hedging reduces the cost of financial distress (Smith and Stulz 1985), firms with high financial leverage would have more incentives to hedge (Nance et al. 1993). Thus, a negative relationship between leverage and forex exposure is anticipated and reported by prior studies (He and Ng 1998). On the contrary, Wei and Starks (2013) argue that although financially distressed firms have incentives to hedge, they have limited ability to employ financial or operational hedges due to high hedging costs and poor credit ratings. As a result, firms with high leverage could hedge less and thus are exposed more. A positive relationship between leverage and forex exposure is also observed by other studies (Chow and Chen 1998; Muller and Verschoor 2007; Hutson and Stevenson 2010; He, Liu, and Zhang 2021).

Similarly, firms with higher growth opportunities have more incentives to hedge to reduce the underinvestment problem (Froot, Scharfstein, and Stein 1993) and should be less exposed. Prior studies have found a weak relationship between firms' growth opportunities and exposure (He and Ng 1998), while several studies have reported a positive relationship contrary to the hedging theory (Aggarwal and Harper 2010; Hutson and Stevenson 2010; Wei and Starks 2013; Hutson and Laing 2014; He, Liu, and Zhang 2021).

Firms' liquidity position serves as a substitute for hedging and firms can reduce the probability of financial distress by holding large liquid assets (Nance et al. 1993). Liquid firms have, therefore, fewer incentives to hedge and are more exposed to forex risk. Prior evidence on the relationship between quick ratio and exposure is mixed. While a few studies report a significant (He and Ng 1998; Krapl 2017) as well as insignificant positive relationship (Hutson and Stevenson 2010; Hutson and Laing 2014; Loriot, Hutson, and Au Yong 2020), a significant negative relationship is also reported (Chow and Chen 1998; Muller and Verschoor 2007).

Asset Utilization

A firm's ability to get adjusted to the dynamic competitive environment determines the vulnerability of its cashflows to exchange rate volatility (Aggarwal and Harper 2010). Therefore, firms with higher asset efficiency should have lower forex exposure (Ye, Hutson, and Muckley 2014; Krapl 2017) as they have natural protection against adverse price changes and the competitive forces of the market.

Profitability

Firms with high profitability would be less venerable to changes in exchange rates as they would have more flexibility in pricing goods and services (Aggarwal and Harper 2010). On the other hand, profitable firms have fewer incentives to hedge as the financial distress cost for them is lower (Muller and Verschoor 2007; Carlos, Sophie, and Tang 2018), resulting in their high forex exposure.

Involvement in international activities

Prior literature has unarguably found that forex exposure increases with their involvement in foreign activities and analysed the effects of foreign assets, foreign sales, or foreign profits on exposure (Jorion 1990; Choi and Prasad 1995; He and Ng 1998). Due to the large multi-country sample of this study, these data are not available. Therefore, following prior studies (Chue and Cook 2008;

Hutson and Stevenson 2010), this study uses the country's openness as a proxy for the firms' involvement in international activities. Firms belonging to a country with high openness are assumed to be involved more in international activities and therefore, are likely to be more exposed.

Industry

Industries differ in terms of pass-through and markups (Allayannis and Ihrig 2001; Bodnar, Dumas, and Marston 2002) or competitive structure (Williamson 2001) and, hence may face different levels of exposure.

Forex exposure during crisis

Prior studies that analyse forex exposure during crisis periods are limited (Aquino 2005; Muller and Verschoor 2007). Although most studies found higher exposure during crisis periods, they have not explored which firm-level factors explain the increased exposure during times of crisis. Firms need to know what specific factors make them more vulnerable to exchange rate fluctuations during periods of crisis.

This study fills this gap and conducts a comprehensive investigation of how firm-level factors affect firms' exposure before and during the COVID-19 crisis.

III. Data, sample, and methods

Sample

First, the top 50 countries in terms of total reported cases of COVID-19 (Zhang, Hu, and Ji 2020; Okorie and Lin 2021) were selected using Worldometer data as of November (Loriot, Hutson, and Au Yong 2020). Next, only those countries were identified that fall under floating or free-floating exchange rate regimes, as classified by the IMF in its 2019 annual report on exchange arrangements and exchange restrictions.[3] This narrows down the country list to 20[4] The sample firms were selected from the major stock market indices of the sample countries (see Table A1-Appendix A) and were sorted into various industry groups based on the FTSE Industry Classification Benchmark. These data are available from Bloomberg. After excluding financial firms and firms with missing observations on stock price data, the final sample consists of 1,577 firms. MSCI developed market and emerging market indices are considered to classify the countries into developed and developing. There are 9 developed countries- Canada, Japan, Italy, Sweden, the UK, the US, Germany, France, Spain, and 11 developing countries- Brazil, China, Chile, Colombia, India, Mexico, Peru, Philippines, South Africa, Turkey, Poland. Table 3 lists the sample countries, the number of firms from each country, and the firms' industry classification.

Models to estimate forex exposure

This section describes the econometric models used to estimate the sample firms' forex exposure. The description starts with the benchmark linear model followed by the asymmetric and non-linear models to have a comprehensive view of exposure.

Linear Model

Following prior studies (Parsley and Popper 2006; Al-Shboul and Anwar 2014), the linear forex exposure of each firm is estimated using the extended version of Jorion's (1990) two-factor model:

$$R_{i,t} = \gamma_{0i} + \gamma_{mi}.R_{m,t} + \gamma_{wmi}.R_{wm,t} + \gamma_{si}.R_{er,t} + \omega_{i,t} \quad (1)$$

with $\omega_{i,t} = \mu_{i,t}\sqrt{\sigma^2{}_{i,t}}$

and $\sigma^2{}_{i,t} = \gamma + \tau_i\sigma^2{}_{i,t-1} + \varphi_i\varepsilon^2{}_{i,t-i}$

where $\sigma^2{}_{it}$ is the conditional variance of residual ω_{it} and γ, τ_j, andφ_k are unknown parameters. μ_{it} denotes error term. If the residuals ω_{it} exhibit conditional heteroscedasticity, a GARCH (1,1) term[5] is added to Equation. (1)[6] Otherwise, the OLS is performed and the standard errors for heteroscedasticity and

[3]This is because the exchange rates in these countries are largely market-determined and do not follow a particular path. As a result, exchange rates are highly volatile in these markets and firms are likely to be exposed more to forex risks.
[4]Based on the suggestion of the anonymous referee, the countries with pegged exchange rate regimes were also considered. However, due to the unavailability of data on IMF's exchange rate indices for most countries, the analysis was conducted only for 3 countries (see Table A5-Appendix A)..
[5]Consistent with (Verschoor and Muller 2007; Bacha et al. 2013) and others.
[6]Equation (1) was also estimated by taking the first-lagged exchange rate changes along with the contemporaneous changes, however, the results were quantitatively similar.

Table 3. Industry classification of sample firms.

Country	No.	Utilities	Tel.	Tech.	Oil & Gas	Industrials	Healthcare	Basic Materials	CS-Media	CS-Retail	CS-Travel & Leisure	CG-Automobiles & Parts	CG-Food & Beverage	CG-Personal & Household
Developed														
Canada	161	13	4	8	26	26	4	44	5	12	5	3	3	8
France	33	2	1	3	1	10	2	1	2	2	1	3	2	3
Germany	22	2	1	2	0	4	4	2	0	0	0	5	2	0
Italy	57	2	3	7	5	17	3	4	0	6	1	0	3	6
Japan	198	5	4	15	3	65	12	30	5	10	9	13	12	15
Spain	24	4	2	3	3	4	2	2	0	1	1	1	1	0
Sweden	19	0	2	2	0	8	1	3	0	1	0	0	0	2
UK	73	5	2	2	2	14	4	11	7	8	6	0	3	9
US	390	28	4	57	26	85	54	17	11	30	22	6	19	31
Developing														
Brazil	37	6	1	1	2	5	2	2	0	9	1	0	5	3
Chile	18	7	1	1	1	1	0	2	0	3	0	0	2	0
China	181	6	1	23	2	48	20	24	4	6	8	8	18	13
Colombia	11	2	1	0	2	3	0	1	0	1	0	0	1	0
India	139	2	2	5	5	7	3	6	10	7	10	23	20	39
Mexico	18	1	1	0	0	6	1	3	0	1	1	0	3	1
Peru	20	4	0	0	1	3	0	7	0	1	0	0	4	0
Philippines	19	4	2	0	0	4	0	1	0	3	2	0	3	0
Poland	12	0	1	0	3	0	0	4	1	1	0	0	0	2
South Africa	73	0	4	3	1	16	4	17	0	16	5	1	6	0
Turkey	72	4	2	4	3	21	2	13	0	5	3	6	5	4
Total	1577	97	39	136	86	347	118	194	45	123	75	69	112	136

autocorrelation are corrected using Newey and West (1987). The rolling regressions over a 36-month window were performed to obtain the coefficients for each firm. R_{it} is the daily return on the firm's stock, R_{mt} is the return on a local market index, R_{wmt} is the return on the MSCI World market index, and R_{ert} is the change in the exchange rate index created by the IMF in nominal terms.

Jorion's (1990) takes returns on the domestic market index along with changes in exchange rates in the regression, while Parsley and Popper (2006) included both local as well as world market returns in their model. They argue that it produces more stable and meaningful estimates because this controls for all domestic as well as global factors that affect the stock returns other than the exchange rates. They used world market returns in the country's local currency. Similarly, Ye, Hutson, and Muckley (2014) also argue that the two-factor model is not ideal for multi-country studies as there may be some confounding world-level macroeconomic factors that may simultaneously affect the stock returns and exchange rates. In the support of these arguments, Chue and Cook (2008) included world market returns instead of the local market index. However, this approach also has some limitations.

Prior research has shown that emerging or frontier markets are less integrated and weakly correlated with world markets (Bekaert 1995). Moreover, for developing countries, local markets are not fully efficient to absorb the effects of all global factors. Following this argument, Ye, Hutson, and Muckley (2014) used emerging market index returns along with world market returns to estimate the emerging market firms' exposure. Since our study is a comprehensive work including both developed and developing countries, the usage of local market returns along with world market returns is believed to be appropriate as it covers all the country-specific as well as worldwide factors that can affect stock returns other than exchange rates[7]

The data for this study is sourced from Bloomberg. An increase in R_{ert} indicates that the local currency of the country appreciates against a group of foreign currencies. A significant regression coefficient γ_{si} indicates that the firm's stock is

[7]To make sure that the findings are not driven by this aspect of the specification, Equation (1) is re-estimated without taking the domestic market returns. The results are quantitatively similar..

affected by exchange rate changes. A positive (negative) value of γ_{si} indicates that the 1% appreciation of the local currency resulted in a higher (lower) stock return for the firm.

The full sample period (Jan. 2017- Feb. 2021) is divided into two sub-periods – A pre-COVID -19 period and a COVID-19 period. The pre-COVID-19 period starts on 3 January 2017, to avoid any overlap with prior crises (Akhtaruzzaman et al. 2020) and continues till the date of the first reported case for each country (Rizwan, Ahmad, and Ashraf 2020; Iyke and Ho 2021). The COVID-19 period ends on the day when the vaccination programme begins in the country (see Table A2-Appendix A for specific time windows). Many recent studies have considered the years 2020 and 2021 as the COVID-19 period for their analysis (Brzeszczyński et al. 2022; Luo 2022; Yu 2022).

Following prior work (Parsley and Popper 2006; Ye, Hutson, and Muckley 2014), the interactive time dummies are added to Equation (1) to determine period-by-period linear exposure:

$$R_{i,t} = \gamma_{0i} + \gamma_{mi}.R_{m,t} + \gamma_{wmi}R_{wm,t} + \sum_{n=1}^{2} \gamma_{si,n}D_{n,t}R_{er,t} + \omega_{i,t} \quad (2)$$

with $\omega_{i,t} = \mu_{i,t}\sqrt{\sigma^2{}_{i,t}}$

and $\sigma^2 i, t = \gamma + \tau_i \sigma^2{}_{i,t-1} + \varphi_i \varepsilon^2{}_{i,t-i}$

where D_n is a dummy variable. $D_1 = 1$ (Pre-COVID -19 period) or zero, otherwise. $D_2 = 1$ (COVID-19 period) or zero, otherwise.

Model for asymmetric and non-linear effects

Most prior literature considers a linear relationship between exchange rates and firms' cash flows, however, a few studies argue that the effect of changes in exchange rates on firms' cash flows is non-linear because firms react to these changes in different ways (Giddy and Dufey 1995). A firm's stock returns may react asymmetrically to currency appreciation and depreciation or to large and small exchange rate changes due to firms' asymmetric pricing-to-market behaviour, hysteresis, and asymmetric hedging (Koutmos and Martin 2003). Koutmos and Martin (2003) propose that exporters with market share objectives exhibit asymmetric pricing-to-market behaviour where they prefer not to increase the foreign currency prices in case of domestic currency appreciation but lower the foreign prices to increase their sales when the domestic currency depreciates. Furthermore, sometimes firms tend to involve in asymmetric hedging in which exporters are inclined to hedge against domestic currency appreciations and may remain unhedged against domestic currency depreciation. These behaviours lead to an asymmetric response of firms' cash flows to exchange rate changes. Although a few studies find empirical support for the non-linear relationship (Bartram 2004), there is a lack of consensus regarding a uniform non-linear functional form that could be used across all firms. The studies have found support for quadratic (Priestley and Ødegaard 2007), cubic (Bartram 2004; He, Liu, and Zhang 2021), and asymmetric effects (Muller and Verschoor 2006a; He, Liu, and Zhang 2021). To analyse the possibility of these effects as well as to provide robustness of the benchmark results of linear exposure, the models to capture asymmetric and non-linear effects are estimated.

Following Koutmos and Martin (2003), the asymmetrical effects of domestic currency appreciation and depreciation on firms' stock returns are estimated using the following model:

$$R_{i,t} = \gamma_{0i} + \gamma_{mi}.R_{m,t} + \gamma_{wmi}.R_{wm,t} + (\gamma_{si} + \gamma_{Dsi}D_t)R_{er,t} + \omega_{i,t} \quad (3)$$

where $D_t = 1$ if R_{ert} <0, or zero otherwise. A statistically significant γ_{Dsi} indicates the presence of an asymmetric exposure of the firm irrespective of the sign of the coefficient. γ_{si} and $\gamma_{si} + \gamma_{Dsi}$ are the measure of the firm's exposure when R_{ert} >0 and R_{ert} <0, respectively. The ratio $(\gamma_{si} + \gamma_{Dsi})/\ \gamma_{si}$ is a measure of the degree of asymmetric response of stock returns to changes in exchange rates.

To examine the non-linear effects of exchange rate changes on stock returns, the cubical function was used which estimates the convex forex exposure (Bartram 2004; Kizys and Pierdzioch 2007). Equation (1) is extended by adding squared and cubic-powered exchange rate returns as follows:

$$R_{i,t} = \gamma_{0i} + \gamma_{mi}.R_{m,t} + \gamma_{wmi}.R_{wm,t} + \gamma_{si}.R_{er,t} + \gamma_{sqi}.{R_{er,t}}^2 + \gamma_{sci}.{R_{er,t}}^3 + \omega_{i,t} \quad (4)$$

A significant value of γ_{sqi} or γ_{sci} indicates the presence of nonlinear exposure of the firm. Equation 3 and 4 are estimated for two sub-periods – the pre-COVID-19 period and the COVID-19 period.

Cross-sectional regression model

This section describes the cross-sectional regression model to analyse the relationship between forex exposure and its determinants in the dynamic context of the COVID-19 crisis. Prior literature argues that firm-level factors can help only in explaining the magnitude rather than the direction of forex exposure and considered the absolute value of exposure in the cross-sectional analysis (Aggarwal and Harper 2010; Bartram 2004; Choi and Prasad 1995; Doukas, Hall, and Lang 2001; Faff and Marshall 2005; Francis et al. 2017; He, Liu, and Zhang 2021; Hutson, Laing, and Ye 2019; Hutson and Laing 2014; Hutson and O'Driscoll 2010; Hutson and Stevenson 2010; Loriot, Hutson, and Au Yong 2020; Ye, Hutson, and Muckley 2014). A high absolute exposure coefficient signifies high sensitivity of stock returns to exchange rates irrespective of the sign of the coefficient. So, taking absolute values in the cross-sectional analysis helps us to easily explain which factors make firms more or less vulnerable to exchange rate changes, in any direction.[8] Consistent with the standard practice, the absolute value of the firms' exposure coefficients estimated from Equation (2) are regressed on several firm-level variables:

$$|\gamma_{si}| = \beta_{0i} + \beta_1 Size_i + \beta_2 Debt_i + \beta_3 PB_i + \beta_4 Quick_i + \beta_5 Turnover_i + \beta_6 OPM_i + \beta_7 Openness_{ji} + \sum_{k=1}^{12} \beta_8^k INDDUM_i^k + \omega_i \quad (5)$$

where i and j subscript denote firm and country. $|\gamma_{si}|$ is the forex exposure coefficient in absolute value. The variable, *Size*, is measured by the natural logarithm of the firm's total assets. Total debt to total assets (*Debt*), quick ratio (*Quick*), and price to book ratio (*PB*) are used to measure variables that proxy for hedging incentives including financial leverage, liquidity, and growth opportunities respectively. Total asset turnover (*Turnover*) and operating profit margin (*OPM*) measure firms' asset utilization and profitability respectively. The trade to GDP ratio (*Openness*) is a country-level variable measuring the involvement of firms in international activities. $INDDUM_i^k$ represents the 12-industry dummy variables with the industry group Industrials as a reference and ω_i is the error term. The definitions of the variables are presented in Table A3-Appendix A. The variables are annual averages calculated over the pre-COVID period (2017–2019) and COVID-19 period (2020) and the cross-sectional regression is run for both periods separately. Equation (5) is estimated using OLS with heteroskedasticity corrected standard errors.

IV. Results and discussion

This section presents the results and analysis of the study. Table 4 shows that the COVID-19 period witnessed a significant change in the firm-level factors from the pre-COVID-19 period. The preliminary evidence based on the univariate tests indicates that the variables differ significantly across the two periods. Specifically, operating profit margins and asset turnover significantly declined for all sample firms witnessing a higher decline for firms from developed markets than in developing countries. On the contrary, the debt level is significantly increased for almost all the firms. The univariate tests comparing the firm-level variables between developed and developing countries (Table A4-Appendix A) suggest that these variables differ significantly between the two groups for both periods. The firms from developed countries are larger and more profitable than emerging market firms. On the other hand, emerging market firms have higher debt than their counterparts.

[8]Theoretically, if an increase in the exchange rate indicates an appreciation of the local currency, importers (exporters) should benefit (harm) from the local currency appreciation and would have a positive (negative) exposure coefficient. However, in practice, the sign of exposure seems to be less distinct for firms that have both foreign currency payables and receivables and is thereby a complex phenomenon to interpret through firm-level determinants (He and Ng 1998; Muller and Verschoor 2006b; Priestley and Ødegaard 2007).

Table 4. Change in the firm-level variables during the COVID-19 period from the pre-COVID-19 period.

Country	Turnover	PB	Quick	Debt	OPM
Developed					
Canada	−0.08***	0.26	−0.02	1.88***	−2.31
France	−0.10***	0.2	0.08	6.34***	−3.44**
Germany	−0.12***	0.02	0.07	3.54***	−5.08**
Italy	−0.13***	−0.08	−0.04	2.16***	−2.2***
Japan	−0.01**	−0.26***	0	0.16	−0.37
Spain	−0.08***	0.69	0.13	3.95**	−5.73**
Sweden	−0.05**	0.35	0.04	0.98	−2.03
UK	−0.14***	−0.06	0.15**	3.33***	−2.29
US	−0.08***	0.47	0	2.65***	−2.60***
Average	−0.08***	0.21	0.06	2.85**	−2.98**
Developing					
Brazil	−0.07	0.06	0.06	2.8	−1.88
Chile	−0.03	−0.45***	0.14	1.720*	4.6
China	−0.04***	2.23***	0.03	0.04	1.64
Colombia	−0.02	−0.05	−0.02	0.62	−1.66
India	−0.05***	−1.03***	0.03	−1.009*	3.46
Mexico	−0.07**	−0.21	−0.01	6.97***	−5.18**
Peru	−0.14***	−0.2	−0.09	0.74	−5.29*
Philippines	−0.11**	−0.42***	−0.01	2.86*	−7.20***
Poland	−0.12**	0.38	−0.28	4.85	−4.03
South Africa	−0.04	−0.48***	0.11	2.26**	2.38
Turkey	−0.09***	1.04**	0.05	0.53	−1.09
Average	−0.07**	0.08*	0.00	2.04**	−1.30**
Grand Average	−0.08**	0.13*	0.02	2.62***	−1.74**

This table reports the average significant difference in the firm-level variables between the two periods. Details of the variable definitions and specific time period windows are provided in Tables A3 and A2 (Appendix-A) respectively. *, ** and *** denote significance levels at 10, 5, and 1% respectively.

Exposure estimates

This section presents the results of the estimation of the exposure coefficients for sample firms by country using different models. All the basic regression assumptions were conformed while estimating the econometric models. Table 5 summarizes estimated linear exposure coefficients, using Equation (2) for the two time periods. The results exhibit a strong presence of significant linear exposure of sample firms. The number of firms that had significant forex exposure was 389 (24%) in the pre-COVID-19 period (Column 1), which got increased to 443 (28%) during the COVID-19 period (Column 6). The prevalence and magnitude of negative exposure are higher than the positive exposure for firms from developed countries and vice versa for developing countries. This indicates that most firms in developed (developing) countries harm(benefit) from the appreciation of the local currency against the foreign currencies. This can be attributed to higher export orientation of firms in developed economies compared to emerging economies which are largely import-oriented (Muller and Verschoor 2007).

The average magnitude of exposure in absolute value for sample firms significantly increased for each country during the COVID-19 period (Columns 4, 9, and 11) indicating that irrespective of the direction, firms' stock returns became more sensitive to exchange rates fluctuations during the period of crisis. The economic significance of the exposure is calculated as a product of the volatility of the exchange rate and the firm's exposure (Columns 5 and 10) which reflects the effect of one standard deviation of exchange rate changes on the firm's stock returns. Overall, firms' stock returns are, on average, 2.4% more affected by exchange rate volatility during the COVID-19 period than in the pre-COVID-19 period. Surprisingly, firms from the developed countries face significantly higher linear exposure than emerging market firms during COVID-19.

Table 6 presents the summary of linear exposure coefficients estimated for each industry. The mean absolute exposure increases significantly in the COVID-19 period from the pre-COVID-19 period for each industry group. The Oil & Gas industry is the most affected during the COVID-19 period, with a significant increase of around 6.4% in average exposure, followed by Travel & Leisure (4.04%) and Utilities (2.88%). This was expected, given the lockdowns and travel restrictions imposed by various countries on these sectors. On the other hand, the industry groups, such as Healthcare, Technology, and Consumer Goods – Food & Beverages, witnessed the smallest increases in their average exposures.

Further, this study estimates the asymmetrical and non-linear exposure of sample firms. Table 7 reports the asymmetric responses of firms' stock returns to currency appreciation and depreciation estimated from Equation 3. The results suggest weaker evidence of asymmetric exposure as compared to linear exposure for both periods consistent with Hutson and Laing (2014). Only 11 and 15% of sample firms have significant asymmetric exposure during the pre-COVID-19 period and the COVID-19 period respectively. Firms in most countries have average negative exposure during the local currency appreciation which might indicate the asymmetric pricing-to-market behaviour of exporters. Overall, the degree of asymmetry is

Table 5. Linear Exposure coefficients.

		Pre-COVID-19 period					COVID-19 period					
Country	No. of firms	No. of firms with significant exposure (±) (1)	Average positive significant exposure (2)	Average negative significant exposure (3)	Average absolute significant exposure (4)	Economic significance of exposure (5)	No. of firms with significant exposure (±) (6)	Average positive significant exposure (7)	Average negative significant exposure (8)	Average absolute significant exposure (9)	Economic significance of exposure (10)	Difference (t-test significance) (11)
Developed												
Canada	161	45(24/21)	0.459	−0.32	0.395	0.005	60(47/13)	2.51	−1.048	2.199	0.032	1.804***
France	33	9(4/5)	0.279	−0.331	0.308	0.001	4(2/2)	0.955	−1.13	1.062	0.006	0.754**
Germany	22	5(2/3)	0.537	−0.589	0.569	0.003	7(1/6)	1.04	−1.711	1.615	0.012	1.046**
Italy	57	11(1/10)	0.724	−1.09	1.063	0.005	20(0/20)	-	−4.143	4.143	0.026	3.080***
Japan	198	41(22/19)	0.324	−0.373	0.347	0.005	60(16/44)	0.507	−0.543	0.534	0.011	0.187*
Spain	24	7(0/7)	-	−1.2	1.204	0.004	9(3/6)	1.935	−2.89	2.573	0.014	1.369**
Sweden	19	5(4/1)	0.264	−0.31	0.273	0.003	3(2/1)	0.843	−0.74	0.809	0.010	0.536**
UK	73	41(31/10)	0.577	−0.367	0.526	0.008	20(15/5)	1.304	0.688	1.150	0.021	0.624***
US	390	84(39/45)	0.355	−0.418	0.389	0.004	130(41/89)	1.126	−1.621	1.465	0.019	1.076***
Total/ Average	977	248(127/121)	0.439	−0.555	0.564	0.004	313(127/186)	1.277	−1.459	1.728	0.017	1.164***
Developing												
Brazil	37	10(5/5)	0.336	−0.278	0.308	0.008	10(9/0)	0.57	-	0.571	0.024	0.263*
Chile	18	3(3/0)	0.173	-	0.173	0.003	1(0/1)	-	−0.682	0.682	0.021	0.509*
China	181	17(11/6)	0.729	−0.78	0.747	0.006	35(20/15)	1.016	−1.281	1.130	0.010	0.383*
Colombia	11	2(2/0)	0.151	-	0.151	0.003	4(1/3)	0.154	−0.527	0.434	0.016	0.283**
India	139	29(20/9)	0.597	−0.637	0.609	0.007	16(7/9)	1.244	−1.429	1.348	0.016	0.739***
Mexico	18	2(2/0)	0.19	-	0.190	0.004	9(6/3)	0.647	−0.273	0.522	0.023	0.332***
Peru	20	5(5/0)	0.546	-	0.546	0.005	3(0/3)	-	−0.586	0.586	0.008	0.04
Philippines	19	1(0/1)	-	−0.319	0.319	0.003	7(1/6)	1.54	−1.92	1.872	0.019	1.553*
Poland	12	1(0/1)	-	−0.497	0.497	0.004	4(2/2)	1.76	−1.01	1.387	0.017	0.890***
South Africa	73	43(32/11)	0.356	−0.519	0.398	0.011	34(30/4)	0.704	−1.002	0.739	0.024	0.341***
Turkey	72	28(17/11)	0.234	−0.239	0.236	0.009	7(5/2)	0.538	−0.752	0.599	0.011	0.363**
Total/ Average	600	141(97/44)	0.368	−0.467	0.379	0.006	126(81/45)	0.908	−0.946	0.897	0.017	0.518**
Grand total/ average	1577	389(224/165)	0.401	−0.516	0.462	0.005	443(208/235)	1.081	−1.189	1.270	0.017	0.808***

This table reports the results from Equation (2). Columns (1) and (6) present the number of firms with significant forex exposure at a 10% or better significance level and firms with positive and negative exposure in the parenthesis. Rest other columns report the magnitude of forex exposure. Column 11 shows a significant difference in the average absolute exposure between the two periods. Details of specific time period windows are provided in Tables A2 (Appendix-A).

Table 6. Industry-Wise linear forex exposure.

	Pre-COVID-19 period			COVID-19 period		
Industry name	No. of firms	No. of firms with significant exposure	Average absolute significant exposure	No. of firms with significant exposure	Average absolute significant exposure	Difference (t-test significance)
Basic Materials	194	65	0.499	56	1.053	0.554***
CG-Automobiles & Parts	69	12	0.399	17	1.158	0.759***
CG-Food & Beverage	112	25	0.374	27	0.938	0.564***
CG-Personal & Household	136	25	0.56	40	1.749	1.189***
CS-Media	45	10	0.449	11	1.351	0.902***
CS-Retail	123	42	0.407	41	1.083	0.676***
CS-Travel & Leisure	75	26	0.472	27	2.38	1.908***
Healthcare	118	24	0.611	23	1.29	0.679***
Industrials	347	79	0.382	93	1.15	0.768***
Oil & Gas	86	19	0.445	34	3.296	2.851***
Technology	136	25	0.62	43	1.077	0.457***
Telecommunications	39	9	0.309	8	0.746	0.437***
Utilities	97	28	0.327	23	1.27	0.943***
	1577	389	0.438	443	1.4	0.962

This table reports the results of the linear model from Equation (2) for each industry. *** denotes significance level at 1%. Tel. is Telecommunications, Tech. is Technology, CS is Consumer services and CG is Consumer Goods. Details of specific time period windows are provided in Tables A2 (Appendix-A).

Table 7. Asymmetric Forex exposure.

		Pre-COVID-19 period				COVID-19 period			
Country	No. of firms	No. of firms with asymmetric exposure (1)	Average significant exposure during appreciation (2)	Average significant exposure during depreciation (3)	Degree of asymmetry (4)	No. of firms with asymmetric exposure (5)	Average significant exposure during appreciation (6)	Average significant exposure during depreciation (7)	Degree of asymmetry (8)
Developed									
Canada	161	14	0.162	−0.201	−1.24	33	−2.00	0.89	−0.44
France	33	7	0.34	−0.47	−1.38	2	0.06	−0.09	−1.55
Germany	22	3	0.69	−0.45	−0.65	3	0.22	−0.33	−1.50
Italy	57	6	1.00	−2.30	−2.29	17	−5.81	3.30	−0.57
Japan	198	31	−0.22	0.19	−0.84	26	−0.23	−0.09	0.39
Spain	24	4	0.11	−0.12	−1.09	1	0.88	−0.99	−1.13
Sweden	19	1	0.27	0.59	2.19	1	0.52	−2.17	−4.19
UK	73	12	0.15	−0.03	−0.21	8	−0.24	0.19	−0.77
US	390	20	0.14	−0.38	−2.68	44	−0.54	0.05	−0.09
Developing									
Brazil	37	1	−0.348	0.482	−1.39	11	−0.435	0.363	−0.83
Chile	18	1	0.161	−0.251	−1.56	4	−0.26	−0.18	0.69
China	181	24	0.708	−0.502	−0.71	30	−0.43	−0.04	0.10
Colombia	11	1	−0.278	0.31	−1.12	3	−0.562	0.276	−0.49
India	139	23	0.05	−0.29	−6.44	19	−1.54	0.76	−0.49
Mexico	18	3	0.10	−0.06	−0.56	2	−0.54	0.23	−0.44
Peru	20	1	0.30	−0.53	−1.73	2	−1.22	1.08	−0.88
Philippines	19	2	−0.16	0.04	−0.26	9	−1.72	−0.46	0.26
Poland	12	1	0.32	−0.55	−1.72	3	0.19	−0.22	−1.16
South Africa	73	6	0.15	0.15	1.01	11	−0.31	0.03	−0.10
Turkey	72	18	0.02	0.14	7.53	10	−0.56	0.52	−0.93
	1577	179	0.1831	−0.211	−0. 756,933	239	−0.73	0.16	−0.71

This table reports the results from Equation (3). Details of specific time windows are provided in Tables A2 (Appendix-A).

not very high in the two time periods and the change in the asymmetry during the COVID-19 period is marginal.

Table 8 shows the number of firms with non-linear forex exposure estimated from Equation 4 during the two sub-periods. The results indicate a weak presence of non-linear exposure during the pre-COVID-19 period as only 18% of sample firms have statistically significant exposure. This number is far lower than the results of the linear exposure of sample firms during that period. This indicates that for only a few firms, large changes in exchange rates have a different effect on firms than small changes.

Table 8. Non-Linear forex exposure.

Country	No. of firms	Pre-COVID-19 period		COVID-19 period	
		No. of firms with significant exposure	Percentage of firms with significant exposure	No. of firms with significant exposure	Percentage of firms with significant exposure
Developed					
Canada	161	7	4.3%	39	24.2%
France	33	2	6.1%	11	33.3%
Germany	22	2	9.1%	7	31.8%
Italy	57	8	14.0%	10	17.5%
Japan	198	48	24.2%	42	21.2%
Spain	24	3	12.5%	12	50.0%
Sweden	19	4	21.1%	5	26.3%
UK	73	31	42.5%	12	16.4%
US	390	30	7.7%	125	32.1%
Developing					
Brazil	37	10	27.0%	16	43.2%
Chile	18	10	55.6%	1	5.6%
China	181	40	22.1%	56	30.9%
Colombia	11	2	18.2%	1	9.1%
India	139	24	17.3%	27	19.4%
Mexico	18	1	5.6%	2	11.1%
Peru	20	3	15.0%	2	10.0%
Philippines	19	3	15.8%	8	42.1%
Poland	12	4	33.3%	5	41.7%
South Africa	73	22	30.1%	15	20.5%
Turkey	72	33	45.8%	8	11.1%
Total	1577	287	18.2%	404	25.6%

This table reports the results from Equation (4). Details of specific time windows are provided in Tables A2 (Appendix-A).

Overall, the findings reveal that forex exposure significantly increased during the COVID-19 period. The stronger evidence of linear exposure was observed than asymmetric or non-linear exposure in both periods in terms of the prevalence and the magnitude of the exposure. Additionally, the average value of adjusted *R*-square is higher for the linear model (0.327) than the asymmetric (0.219) and the non-linear model (0.217). Therefore, for further analysis, we consider linear exposure coefficients.

Determinants of exposure

Table 9 summarizes the results of equation 5 for the full sample (specifications 1, 4, and 5) and two subsamples from developed (specifications 2, 6, and 7) and developing countries (specifications 3, 8, and 9). The analysis is performed separately for the pre-COVID-19 period and the COVID-19 period to analyse whether the relationship between firm-level factors and forex exposure changes during the period of the crisis.

In the pre-COVID-19 period, the relationship between forex exposure and its determinants is mostly consistent with prior literature. As per the expectation, the effects of the variables *Turnover* and *OPM* on forex exposure are significantly negative in all the specifications. These results support our notion that firms with high asset efficiency or high operating profits are naturally able to adjust to the dynamic competitive forces of the market and thus are less exposed to forex risk (Aggarwal and Harper 2010).

In terms of the variables that proxy for hedging incentives, the results are mixed. Consistent with optimal hedging theories, the relationship between leverage (*Debt*) and exposure is negative indicating that firms with high financial leverage hedge more to reduce their financial distress costs (He and Ng 1998; Nance et al. 1993). Contrary to hedging theory, the relationship between firms' growth opportunities (*PB*) and forex exposure is significantly positive in all the specifications. These findings are, however, supported by prior evidence (Hutson and Stevenson 2010; He, Liu, and Zhang 2021) indicating that despite the higher incentives to hedge, growth firms may choose not to do so. Krapl (2017) argues that high-growth firms may prioritize market share objectives over forex risk management and thereby are exposed more. The effect of liquidity (*Quick*) on exposure remains

Table 9. The relationship between forex exposure and its firm-level determinants for the pre-COVID-19 period and the COVID-19 period.

Dependent variable: $\lvert\gamma_{si}\rvert$	Pre-COVID-19 period			COVID-19 period					
	Full Sample	Developed countries	Developing countries	Full Sample		Developed countries		Developing countries	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Firm-level variables									
Size	−0.005	−0.024*	0.031**	−0.004	−0.001	−0.081***	−0.077***	0.042**	0.009*
	(0.008)	(0.013)	(0.014)	(0.012)	(0.012)	(0.021)	(0.021)	(0.003)	(0.013)
Debt	−0.002**	−0.001*	−0.003***	0.003**	0.002**	0.005***	0.004**	0.002**	0.001**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.001)	(0.001)
PB	0.006**	0.006*	0.003*	0.005*	0.004	0.001*	0.001	0.004	0.003
	(0.003)	(0.003)	(0.005)	(0.003)	(0.003)	(0.004)	(0.004)	(0.003)	(0.003)
Quick	0.003	0.005	0.018	0.027**	0.021*	0.011	0.010	−0.017	−0.025
	(0.011)	(0.012)	(0.033)	(0.012)	(0.012)	(0.017)	(0.016)	(0.015)	(0.016)
Turnover	−0.025*	−0.053*	−0.010***	−0.111***	−0.117***	−0.191***	−0.138***	−0.005**	−0.022*
	(0.014)	(0.03)	(0.022)	(0.035)	(0.038)	(0.053)	(0.060)	(0.023)	(0.033)
OPM	−0.003*	−0.003**	−0.005**	−0.005***	−0.003***	−0.011***	−0.005***	−0.010**	−0.009*
	(0.001)	(0.002)	(0.002)	(0.001)	(0.001)	(0.002)	(0.002)	(0.001)	(0.001)
Openness	0.001*	0.002***	0.006***	0.003*	0.003*	0.002***	0.002***	0.009**	0.010**
	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)	(0.003)	(0.003)
Debt interactions									
Debt*Turnover					−0.004**		−0.005**		−0.003**
					(0.002)		(0.003)		(0.002)
Debt*OPM					−0.025**		−0.021**		−0.010*
					(0.003)		(0.003)		(0.003)
Constant	0.422***	0.639***	0.651***	0.645***	0.545***	0.751***	1.326***	0.495***	0.444**
	(0.103)	(0.155)	(0.181)	(0.018)	(0.126)	(0.03)	(0.227)	(0.022)	(0.133)
Observations	1577	977	600	1577	1577	977	977	600	600
Industry dummies	yes	yes	yes	yes	yes	yes	yes	yes	yes
Countries	20	9	11	20	20	9	9	11	11
Adjusted *R*-square	0.015	0.213	0.202	0.043	0.126	0.10	0.200	0.09	0.100
adjusted *R*-square					0.083**		0.10***		0.01**

This table presents the results from equation (5). The dependent variable is the absolute value of the firms' forex exposure estimated from Equation (2). The definitions of the variables are presented in Table A3-Appendix A. All independent variables are taken as yearly averages over the pre-COVID-19 period (2017–2019) and the COVID-19 period (2020). Robust standard errors corrected for heteroskedasticity are reported in parenthesis. ***,** and * denote significance level at 1%, 5% and 10% respectively.

insignificant in all the specifications with the expected direction of the relationship (Hutson and Stevenson 2010).

The effect of firm size (*Size*) on forex exposure is contrasting for the two sub-samples from developed and developing countries. As discussed earlier, based on hedging theories, the relationship between firm size and exposure can be positive or negative. For the firms from developed countries, firm size has a negative effect on exposure supporting the argument that larger firms that have economies of scale in hedging costs are more likely to hedge than smaller firms. On the contrary, for the sub-sample of firms from developing countries, firm size and exposure are positively related. This can be explained by the counterargument that smaller firms hedge more because their financial distress cost is higher than larger firms. This indicates that hedging benefits outweigh the costs for larger firms in developed markets. On the other hand, hedging is more beneficial for smaller firms in developing countries due to their high debt. Sample data (Table A4-Appendix A) also support that firms in developed countries are significantly bigger and have lower financial leverage than firms from developing countries. Finally, the country's openness is positively related to exposure as expected.

Next, the results from the COVID-19 period are quite different from the primary results in terms of their strength, significance, and direction. Noticeably, in contrast to the previous results, firms with high financial leverage are positively associated with forex exposure during the period of COVID-19. This indicates that despite high incentives for hedging, financially distressed firms are less likely to hedge during periods of crisis and therefore, are more

Table 10. The dynamic Panel GMM results.

Dependent variable: $\|\gamma_{si}\|$			
	Full Sample (1)	Developed countries (2)	Developing countries (3)
Lagged Dependent	0.001* (0.003)	0.004 (0.074)	0.011* (0.003)
Firm-level variables			
Size	−0.002 (0.011)	−0.071*** (0.021)	0.009 (0.013)
Debt	−0.002** (0.001)	−0.004** (0.002)	−0.001* (0.001)
PB	0.004 (0.003)	0.001 (0.004)	0.003 (0.003)
Quick	0.009 (0.012)	0.009 (0.016)	0.028 (0.017)
Turnover	−0.094** (0.037)	−0.098* (0.059)	−0.010* (0.036)
OPM	−0.005*** (0.001)	−0.003** (0.002)	−0.001* (0.001)
Openness	0.005* (0.001)	0.002*** (0.001)	0.009** (0.003)
Crisis dummy and interaction variables			
CRISIS	0.094** (0.027)	0.090* (0.059)	0.110* (0.036)
CRISIS*Size	−0.001 (0.022)	−0.088*** (0.022)	0.055** (0.002)
CRISIS*Debt	0.005** (0.001)	0.045*** (0.002)	0.022** (0.001)
CRISIS*PB	0.003* (0.001)	0.004* (0.001)	0.003 (0.002)
CRISIS*Quick	0.026** (0.011)	0.010 (0.011)	0.011 (0.011)
CRISIS*Turnover	−0.112*** (0.020)	−0.120*** (0.013)	−0.020** (0.010)
CRISIS*OPM	−0.009*** (0.001)	−0.012*** (0.002)	−0.010** (0.001)
CRISIS*Openness	0.003* (0.001)	0.002*** (0.001)	0.009** (0.003)
CRISIS*Debt*Turnover	−0.014** (0.002)	−0.055** (0.003)	−0.023** (0.002)
CRISIS*Debt*OPM	−0.020** (0.001)	−0.022** (0.003)	−0.012* (0.003)
Debt*Turnover	−0.014** (0.007)	−0.008* (0.019)	−0.010* (0.016)
Debt*OPM	−0.002 (0.061)	0.07 (0.031)	−0.009 (0.013)
Constant	0.561*** (0.125)	1.216*** (0.226)	0.454** (0.134)
Observations	6308	3908	2400
Industry dummies	yes	yes	yes
Countries	20	9	11
Year fixed effects	yes	yes	yes
Country fixed effects	yes	yes	yes
AR(2)	0.77	0.72	0.79
Sargan Test (*p*-value)	0.76	0.74	0.71

This table presents the results from the dynamic panel GMM Model. The dependent variable is the absolute value of the firms' forex exposure. The definitions of the variables are presented in Table A3-Appendix. Equation (1) was estimated to obtain the firms' exposure coefficient for each year of the sample period 2017–2020. The dummy variable, *CRISIS*, is having a value of 1 for the year 2020, or zero otherwise. One-year lagged values and the difference of the regressors are used as instruments. AR(2) is the *p*-value of the second-order autocorrelation test. Sargan test reports the *p*-value of the over-identifying restrictions. Standard errors are reported in parenthesis. ***,** and * denote significance level at 1%, 5% and 10% respectively.

exposed to currency fluctuations. The firms have a motive to hedge only when the costs of hedging do not exceed the benefits (Nance et al. 1993). Since hedging becomes costlier during crisis periods in comparison to normal periods due to high exchange rate volatility, firms are less motivated or less capable of hedging.

These results complement a strand of prior literature to some extent. Contrary to theory, Allayannis and Ofek (2001), Hutson and Laing (2014), and others found that firms with high debt are less likely to use derivatives. A positive relationship between forex exposure and financial leverage was proposed by Wei and Starks (2013) arguing that sometimes despite the high incentives to hedge, financially distressed firms are not motivated to hedge as they have limited ability to employ financial or operational hedges due to high hedging costs.

The relationship between forex exposure and the rest of the other variables is similar in terms of sign but exhibits variability in terms of strength and significance.

The coefficients of *OPM* and *Turnover* become higher in magnitude and more significant indicating that firms with lower asset efficiency and lower profitability became more sensitive to the changes in the exchange rate during the COVID-19 period. The relationship between *Quick* and forex exposure becomes significant with the same positive sign indicating that highly liquid firms that have fewer incentives to hedge become highly exposed during the crisis. The effect of firm size on exposure is similar in sign but becomes stronger in magnitude. High-growth firms remain significantly positively exposed to exchange rate changes.

Overall, the results provide enough evidence that the effect of firm-level factors on forex exposure changes during periods of crisis.

Financial leverage and forex exposure-interaction analysis

Contrary to theory, the positive relationship between leverage and exposure found during the COVID-19 period is justified based on the argument that firms hedge less as they have limited ability to hedge due to high hedging costs. However, more profitable and efficient firms would have less difficulty in bearing high hedging costs during a financial crisis. Therefore, it can be argued that firms with high leverage can employ hedging if they have high profits or high asset turnover that help them to cover hedging costs and consequently, can limit their forex exposure. To examine how financial leverage affects the exposure of firms with high profits or high turnover, the interaction analysis is conducted in Table 9 (specifications 5,7, and 9). Debt interaction terms, *Debt*Turnover* and *Debt*OPM* are added to equation (5) to investigate the joint impact of leverage-profitability and leverage-turnover on firms' exposure. The analysis is conducted for full sample and two sub-samples during the period of COVID-19.

The results reveal that, during the COVID-19 period, higher financial leverage exhibit lower sensitivity to exchange rate changes for firms with higher profitability or efficiency. The results are robust across all the specifications.

Robustness analysis

To analyse the effects of firm-level determinants on forex exposure in a dynamic framework, the dynamic panel GMM model (Arellano and Bond 1991) is estimated. This technique controls for endogeneity by incorporating the lags of changes in the dependent variable and the endogenous independent variables. To construct panel data, Equation (1) was re-estimated to obtain the firms' exposure coefficient for each year of the sample period 2017–2020. To investigate whether the relationship between firm-level variables and forex exposure changed during the crisis, we add a dummy variable, *CRISIS*, (having a value of 1 for the year 2020, or zero otherwise) to the model and interact it with the variables. Table 10 presents the results of the dynamic panel GMM model which are consistent with the primary results of the study.

V. Implications and conclusion

This study conducts a comprehensive examination of forex exposure and its determinants for a sample of 1577 firms sourced from 20 countries before and during the COVID-19 crisis. The results provide evidence of a significant rise in the linear forex exposure of firms across industries during the COVID-19 period. This study offers useful findings regarding the determinants of exposure during crisis periods and reveals that the effect of firm-level variables on exposure varies across crisis and non-crisis periods. Firms with high asset turnover or large operating profits remain less exposed during times of stress since they can get adjusted to the dynamic competitive environment. On the contrary, firms with high leverage become highly exposed to forex risk during crisis periods. However, they can limit their exposure if they have large operating profits or effective asset utilization which can help them to bear heavy hedging costs.

These findings provide important implications for firms, investors, and policymakers. Effective asset utilization and operational profitability are the keys for firms to limit their exposure to undue currency fluctuations during times of crisis. Since firms are less motivated or are less able to hedge during the crisis periods due to high hedging costs, the management should think about introducing internal policies that would encourage managers to hedge and would help them to reduce the hedging costs. This study is limited due to the lack of availability of data on hedging activities of sample firms. Future research could use firm-level data on hedging instruments to examine the effectiveness of firms' risk-management policies during the period of crisis.

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Appendix A

Table A1. Descriptive Statistics of the stock market returns and changes in the exchange rate indices for sample countries.

	Bloomberg Tickers		Pre-COVID-19 period				COVID-19 period			
Country	Market Index	Exchange rate index	Average $R_{m,t}$	S.D.	Average $R_{er,t}$	S.D.	Average $R_{m,t}$	S.D.	Average $R_{er,t}$	S.D.
Brazil	IBOV Index	BISBBRN index	0.08%	1.22%	−0.02%	0.74%	−0.07%	3.42%	−0.12%	1.23%
Canada	SPTSX Index	BISBCAN index	0.01%	0.53%	0.00%	0.37%	−0.02%	2.58%	−0.01%	0.42%
Chile	IPSA Index	BISBCLN index	0.02%	0.87%	−0.02%	0.57%	−0.10%	2.58%	0.06%	0.87%
China	SHSZ300 Index	BISBCNN index	0.03%	1.09%	0.00%	0.22%	0.09%	1.59%	0.00%	0.27%
Colombia	COLCAP Index	BISBCON index	0.03%	0.69%	−0.02%	0.60%	−0.24%	2.77%	−0.03%	1.07%
France	CAC Index	BISBEUN index	0.03%	0.79%	0.00%	0.11%	−0.13%	2.38%	0.01%	0.16%
Germany	DAX Index	BISBDEN index	0.02%	0.84%	0.00%	0.14%	−0.03%	2.42%	0.02%	0.21%
India	NSE500 Index	BISBINN index	0.04%	0.76%	0.00%	0.11%	−0.04%	2.25%	0.02%	0.16%
Italy	FTSEMIB Index	BISBITN index	0.02%	0.98%	−0.01%	0.31%	−0.12%	2.66%	−0.03%	0.33%
Japan	NKY Index	BISBJPN index	0.03%	0.94%	0.00%	0.13%	−0.04%	1.90%	0.02%	0.18%
Mexico	MEXBOL Index	BISBMXN index	−0.01%	0.84%	0.01%	0.41%	−0.09%	1.74%	0.02%	0.62%
Peru	SPBLPGPT Index	BISBPEN index	0.04%	0.66%	0.01%	0.65%	−0.08%	1.79%	−0.08%	1.25%
Philippines	PCOMP Index	BISBPHN index	0.02%	0.92%	0.00%	0.25%	−0.18%	2.45%	−0.01%	0.40%
Poland	WIG20 Index	BISBPLN index	0.01%	0.76%	−0.01%	0.27%	−0.13%	2.38%	0.01%	0.30%
South Africa	JALSH Index	BISBZAN index	0.02%	0.84%	0.01%	0.26%	0.00%	2.23%	0.00%	0.36%
Spain	IBEX Index	BISBESN index	0.02%	0.82%	−0.01%	0.82%	−0.03%	2.42%	−0.10%	0.95%
Sweden	OMX Index	BISBSEN index	0.02%	0.82%	−0.01%	0.34%	−0.01%	2.11%	0.03%	0.37%
Turkey	XU100 Index	BISBTRN index	0.05%	1.22%	−0.07%	1.12%	−0.05%	1.84%	−0.13%	0.52%
UK	UKX Index	BISBGBN index	0.01%	0.69%	0.00%	0.42%	−0.14%	2.15%	−0.01%	0.53%
US	SPX Index	BISBUSN index	0.05%	0.79%	0.00%	0.28%	0.02%	2.62%	0.00%	0.38%

Table A2. tc

Country	Arrival date of first COVID-19 case	Vaccination date	Pre-COVID-19 period	COVID-19 period
Brazil	February 25, 2020	January 17, 2021	Jan 3, 2017 to Feb 25, 2020	Feb 26, 2020 to Jan 17, 2021
Canada	January 25, 2020	December 14, 2020	Jan 3, 2017 to Jan 25, 2020	Jan 26, 2020 to Dec 14, 2020
Chile	March 3, 2020	February 7, 2021	Jan 3, 2017 to Mar 3, 2020	Mar 4, 2020 to Feb 7, 2021
China	December 31, 2019	December 15, 2020	Jan 3, 2017 to Dec 31, 2019	Jan 1, 2020 to Dec 15, 2020
Colombia	March 6, 2020	February 17, 2021	Jan 3, 2017 to Mar 6, 2020	Mar 7, 2020 to Feb 17, 2021
France	January 24, 2021	December 27, 2020	Jan 3, 2017 to Jan 24, 2020	Jan 25, 2020 to Dec 27, 2020
Germany	January 27, 2020	December 26, 2020	Jan 3, 2017 to Jan 27, 2020	Jan 28, 2020 to Dec 26, 2020
India	January 27, 2020	January 16, 2021	Jan 3, 2017 to Jan 27, 2020	Jan 28, 2020 to Jan 16, 2021
Italy	January 31, 2020	December 27, 2020	Jan 3, 2017 to Jan 31, 2020	Feb 1, 2020 to Dec 27, 2020
Japan	January 16, 2020	February 17, 2021	Jan 3, 2017 to Jan 16, 2020	Jan 17, 2020 to Feb 17, 2021
Mexico	February 27, 2020	December 24, 2020	Jan 3, 2017 to Feb 27, 2020	Feb 28, 2020 to Dec 24, 2020
Peru	March 6, 2020	February 9, 2021	Jan 3, 2017 to Mar 6, 2020	Mar 7, 2020 to Feb 9, 2021
Philippines	January 22, 2020	March 1, 2021	Jan 3, 2017 to Jan 22, 2020	Jan 23, 2020 to Mar 1, 2021
Poland	March 4, 2020	December 27, 2020	Jan 3, 2017 to Mar 4, 2020	Mar 5, 2020 to Dec 27, 2020
South Africa	March 1, 2021	February 17, 2021	Jan 3, 2017 to Mar 1, 2020	Mar 2, 2020 to Feb 17, 2021
Spain	January 31, 2020	December 27, 2020	Jan 3, 2017 to Jan 31, 2020	Feb 1, 2020 to Dec 27, 2020
Sweden	January 31, 2020	December 27, 2020	Jan 3, 2017 to Jan 31, 2020	Feb 1, 2020 to Dec 27, 2020
Turkey	March 11, 2020	January 14, 2021	Jan 3, 2017 to Mar 11, 2020	Mar 12, 2020 to Jan 14, 2021
UK	January 31, 2020	December 8, 2020	Jan 3, 2017 to Jan 31, 2020	Feb 1, 2020 to Dec 8, 2020
US	January 20, 2020	December 14, 2020	Jan 3, 2017 to Jan 20, 2020	Jan 21, 2020 to Dec 14, 2020

Table A3. Variable definitions.

Variable	Definition
Size	Natural logarithm of the firms' total assets
Debt	Total long term debt/Total assets
PB	Price-book ratio=Market price per share/Book value per share
Quick	Quick ratio=(Current assets-inventory)/current liabilities
Turnover	Total asset turnover = Sales/Total assets
OPM	Operating Profit Margin=Operating profit/Sales
Openness	(Exports+Imports)/GDP

Table A4. Univariate analysis.

	Pre-COVID-19 period			COVID-19 period		
	Developing	Developed	Difference (*t*-test significance)	Developing	Developed	Difference (*t*-test significance)
No. of firms	600	977		600	977	
PB	4.4565	4.4309	−0.025	4.945	4.640	−0.305
Quick	1.1620	1.2458	0.083	1.197	1.264	0.066
Debt	27.4930	23.6330	4.139***	29.282	24.808	5.526***
Turnover	0.9078	0.7907	−0.117***	0.850	0.720	−0.129***
OPM	10.6799	8.9367	−1.743***	11.473	6.738	−4.734***
Size	7.7730	9.4041	1.631***	7.962	9.571	1.608***

Table A5. Linear Exposure.

		Pre-COVID-19 period			COVID-19 period		
Country	No. of firms	% of firms with significant exposure	Average positive significant exposure	Average negative significant exposure	% of firms with significant exposure	Average positive significant exposure	Average negative significant exposure
Argentina	14	5%	0.423	−0.323	7%	1.231	−1.54
Saudi Arabia	166	25%	0.788	−0.221	29%	2.955	−1.342
Singapore	21	12%	0.212	−0.564	15%	1.22	−1.995